

PARTS LIST

Semiconductors:

BOARD1	- Arduino Uno R3
T1	- BC547 npn transistor
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 220-ohm
R2	- 470-ohm
R3, R11	- 1-kilo-ohm
R4	- 4.7-kilo-ohm
R5, R7, R9, R10	- 10-kilo-ohm
R6	- 100-kilo-ohm
R8	- 100-ohm

Miscellaneous:

CON1	- 2-pin terminal connector
PZ1	- Piezo buzzer
S1-S8	- Tactile switch
LCD1	- 16x2 LCD module
	- 9V battery
	- 9V DC adaptor

buzzer's negative terminal to the collector of T1 and positive terminal to 9V battery.

Software

Circuit operation is done using the software program (tone_generator.ino) loaded into the internal memory of Arduino Uno R3. The program implements all required functionalities including handling user inputs with the help of switching interface. The program is written in Arduino programming language. Arduino IDE 1.6.4 is used to compile and upload the program (called sketch).

The sketch begins by importing LiquidCrystal.h and MyTones.h libraries, so you can reference various items from these later on in the code.

Next, the code starts by defining a name for S8, inputs versus outputs, and setting the frequency values for each of the seven pushbutton keys (switches S1 through S7).

A new tab is created for each individual tone in an effort to keep the code organised and easier to understand. If you wish to delete or add more tones, make sure to change the code in the main program/sketch as well.

Main loop. The program begins with the LED1 turned off, and all tone keys (S1-S7) active. Once the loop detects that S8 has been pressed, it disables S1 through S7, and turns the LED1 on to indicate that you are in Menu mode.

Accessing the menu. Press Menu

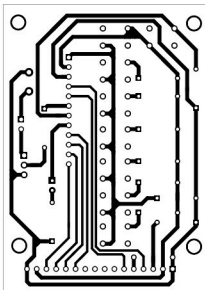


Fig. 3: Actual-size PCB layout of tone generator system

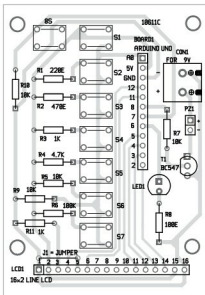


Fig. 4: Components layout of the PCB

button to cycle through the present tones, and press any key (S1 through S7) to start playing the tone displayed on the LCD1 screen.

The LCD1 screen will show the tone that is currently playing and then return to the same point in the menu once the tone has finished playing. From there, you can either replay or continue to play next tone available in the list. Double-click S8 to restart the menu from the beginning. Here, we have created five mu-

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

sical tones: Super Mario Bros Theme song, Super Mario Bros Underworld song, Jeopardy Theme song, and the fourth and fifth are same modified version of Jeopardy Theme song generated for you to modify.

To return to tone keys, cycle through the remaining tones in the menu. Once LCD1 screen is blank and LED1 is off, press each key (S1 to S7) to generate unique beep tones.

Construction and testing

An actual-size, single-side PCB for the electronic tone generator is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit, enclose it in a suitable box.

Fix CON1, LED1 and S1 through S8 on the front panel of the box. Connect the 9V using a 9V adaptor to Arduino Uno board and 9V battery for PZ1.

Testing procedure

After assembling the circuit, upload the program into Arduino Uno board. Connect 9V adaptor to Arduino Uno. LCD1 is blank and LED1 remains off. Tone switches produce individual tones when each switch is pressed momentarily. Press S8 momentarily and tone switches will get disabled.

When you press S8 again, LCD1 shows the message: "Which song would you like to play?" When you press any tone key, the first tone starts playing.

On further pressing S8, the next tone will be displayed on the LCD1. When you press any tone key again, the second tone starts playing. Thereafter, repeat the same process till the last tone is played. Press S8 again and the LCD1 will be blank. **EFY**



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